Diffusion Model for Inverse Design of Metamaterials
CS 598 SciML

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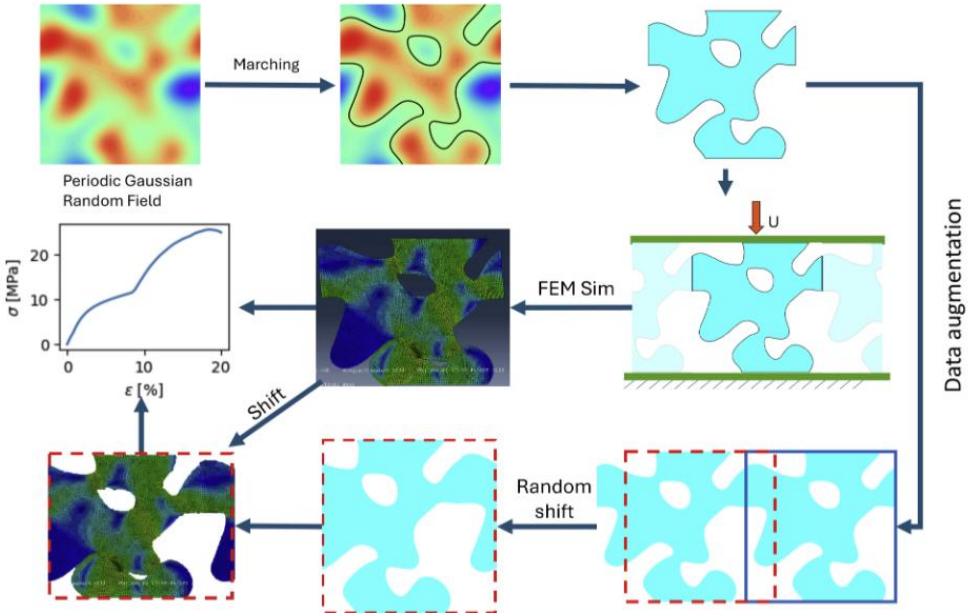
Introduction

- The inverse design of metamaterial unit cells aims to **generate geometries that achieve a target stress–strain response**.



Diffusion Model

Diffusion models are **generative models that learn to generate data by reversing a gradual noise process**.

Forward process: progressively corrupt data with Gaussian noise.

Reverse process: train a neural network to remove the noise step-by-step to reconstruct the data.

Starting from real data x_0 (e.g., an SDF geometry), noise is gradually added over T steps.

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I\right)$$

where

- x_t = noisy sample at step t
- β_t = variance schedule
- I = identity matrix

4. Generation Procedure

To generate a sample:

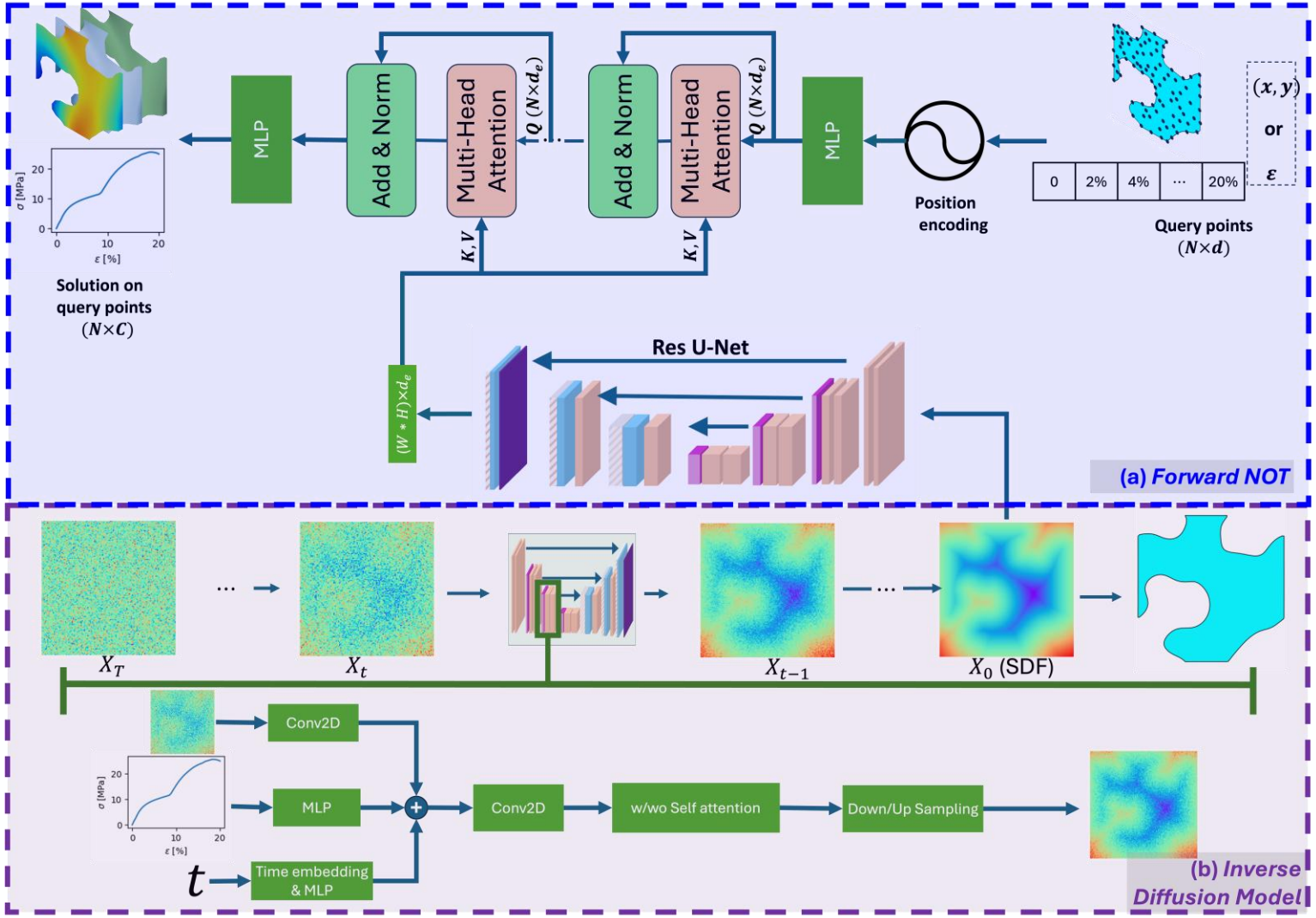
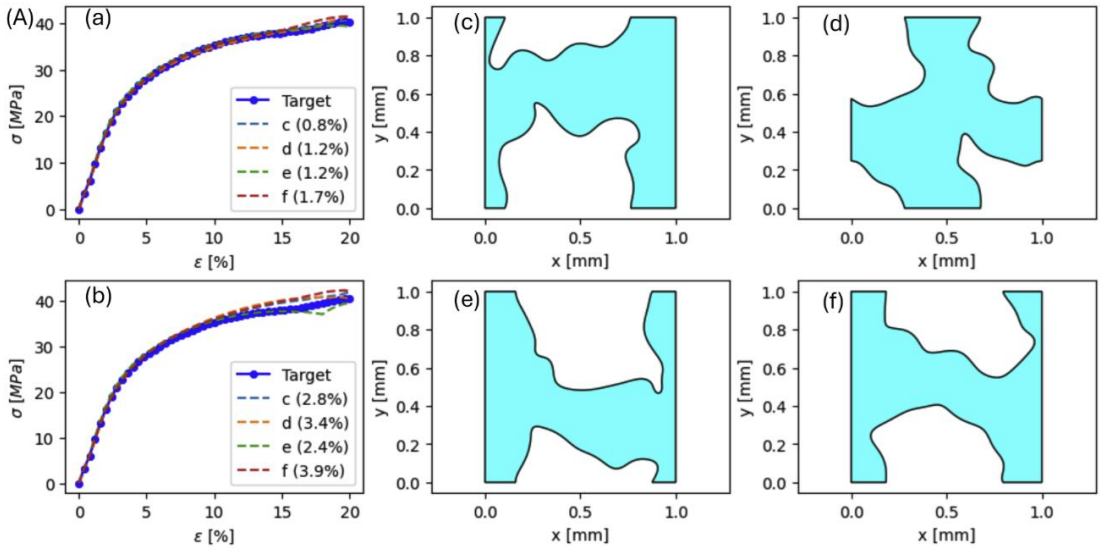
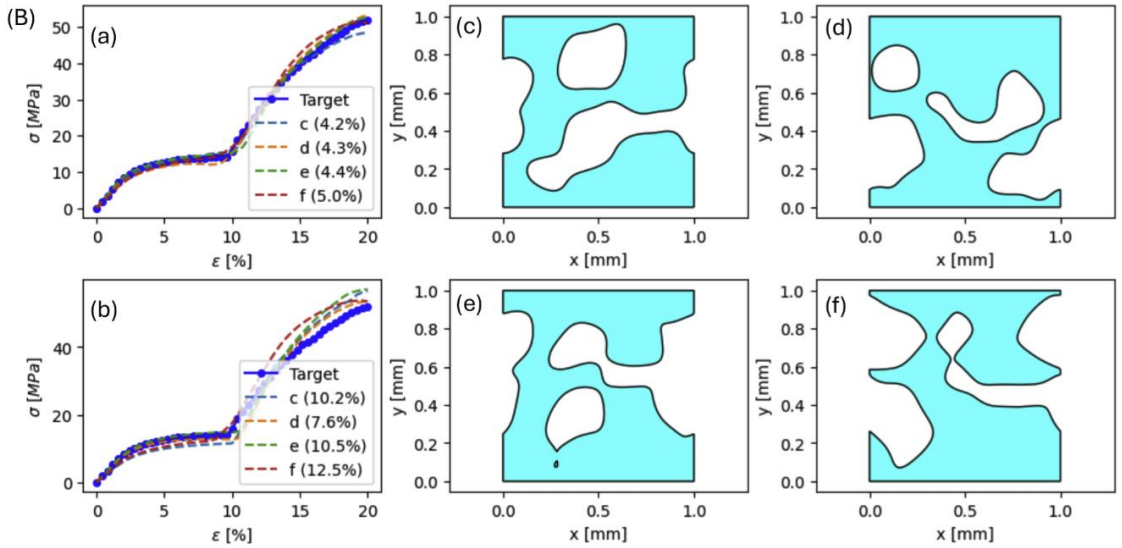
1. Sample Gaussian noise

$$x_T \sim \mathcal{N}(0, I)$$

2. Apply the learned reverse steps

$$x_T \to x_{T-1} \to \ldots \to x_0$$

Model Architecture



Diffusion Model Results – Two Sample Geometry Generation